Paper I

1. Find Voltage across R and C in the following circuits.

a. In a given RC circuit find the voltage across C and R?

b. In a given CR circuit find the voltage across R and C ?;

2. For the given _expression Y=A'B'C+A'BC+AB'C+ABC+ABC' realize using the following

a. 2 input and 3input NAND gate

b. 2 input and 3 input NOR gate

c. AND,OR, INVERTER.

d. INVERTER;

3. What is the importance of scan in digital system.;

4.Given A XOR B =C, such that prove the following

a. B XOR C =A

b. A XOR BXOR C=0;

5. Construct an input test pattern that can detect the result E stuck at 1 in the ckt below

NAND (A,B)->E, NAND(C,D)->F

AND(E,F)->A.

6. In a given opamp ckt input offcet is 5mv,volatage gain =10,000,vsat=+-15v

such that find the output voltage .

7. Draw the p side equation of the circuit.(I am not sulre about it)

8. Make a JK FF using a D FF and 4->1 MUX.

9.Use 2->1 MUX to implement the following _expression

Y=A+BC'+BC(A+B).

10.For the following ckt what is the relation between fin and fout.?

the D FF use +ve edge triggered and have a intial value is 0

CLK->two DFFs with complementing (i.e one DFF have CLK and other one have

Complement of it),inputs of DFF is same and output of DFFs is given to NOR

Gate and output of NOR gate is feedback to the two DFFs.

11. Design a asyncronous circuit for the following clk waveforms.

CLK->thrice the CLK period->half the period of input.

12. What is the setup time and hold time parameters of the FF, what happens if we are not consider it in designing the digital ckt.

13. Given two DFF A,B ones output is the input of other and have the common clock.

Fmax if A and B are +ve edge triggered, if A is+ve edge triggered ,B is -ve edge triggered what is the Fmax relation to previous Fmax relation…

14. What are the FIFOS .? give some use of FIFOS in design.

Paper II

1. What is FIFO ? where it is used?
2. what is set-up and hold time?
3. Two +ive triggered FFs are connected in series and if the maximum frequency that can operate this circuit is Fmax. Now assume other circuit that has +ive trigger FF followed by –ive trigger FF than what would be maximum frequency in terms of the Fmax that the circuit can work?
4. layout of gates were shown and u have to identify the gates (NAND & NOR gates)
5. make a JK FF using a mux(4:1) and a FF.
6. the waveform of clk, i/p and o/p were shown and u have to make a seqential circuit that should satisfy the required waveform.
7. resistor is connected in series with capacitor and the input is dc voltage. Draw the waveform across the capacitor and resistor.
8. two FFs, one is –ive triggered and other is +ive triggered are connected in parallel. The 2 i/p NAND gate is has the i/ps from the q_out of both the FFs and the output of the NAND gate is connected with the I/p of both FFs . Find the frequency of the output of the NAND gate w.r.t clk.

Interview questions (face to face discussion)

1. Draw the circuit for inverter. How does it work?
2. If the pmos and nmos is changed in the inveretr, how does it behave?
3. Design flow for ASICs and FPGA. what are the difference between the ASICs and FPGA?where do u use ASIC and where u use FPGA?
4. What is floorplanning?
5. What do u mean by technology file used in the synthesis or optimization for the circuit (netlist)? What is the difference in the technology files used for the ASICs and FPGAs based designing?
6. Using a FF and gates. Make a memory (i.e include RD, WR etc.)
7. If the setup & hold time gets violated than what u 'll do to remove it?
8. What is clock skew? How u 'll minimize it?
9. What is clock tree? How it looks like? Concept behind that.
10. What about the Vdd and Gnd lines ? does one Vdd and Gnd pins will be sufficient for the chip. What will be the effect of using single Vdd and Gnd pins in the chip?
11. What is voltage refernce circuit? What is bandgap? How does it work?
12. what is FIFO? How does it work? Draw the circuit of FIFO of 1-bit and 4memory location deep? What would happen if memory is full and again u try to write in FIFO? What u 'll do to overcome this problem? Which one would be more easier to implement :- either dropping the packet, when the FIFO is full or pushing the data of FIFO every time. And why ?

The following questions are used for screening the candidates during the prescreening interview. The questions apply mostly to fresh college grads pursuing an engineering career at Intel.

COMPUTER ARCHITECTURE QUESTIONS

1. For a single computer processor computer system, what is the purpose of a processor cache and describe its operation?

2. Explain the operation considering a two processor computer system with a cache for each processor.

What are the main issues associated with multiprocessor caches and how might you solve it?

3. Explain the difference between write through and write back cache.

4. Are you familiar with the term MESI?

5. Are you familiar with the term snooping?

STATE MACHINE QUESTIONS

1. Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.

2. In what cases do you need to double clock a signal before presenting it to a synchronous state machine?

SIGNAL LINE QUESTIONS

1. You have a driver that drives a long signal & connects to an input device. At the input device there is either overshoot,

undershoot or signal threshold violations, what can be done to correct this problem?

VALIDATION QUESTIONS:

What are the total number of lines written in C/C++? What is the most complicated/valuable program written in C/C++?

What compiler was used?

Have you studied busses? What types?

Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an

instruction in a 5 stage machine? What is the throughput of this machine ?

How many bit combinations are there in a byte?

What is the difference between = and == in C?

Are you familiar with VHDL and/or Verilog?

MEMORY, I/O, CLOCK AND POWER QUESTIONS

1. What types of CMOS memories have you designed? What were their size? Speed? Configuration Process technology?

2. What work have you done on full chip Clock and Power distribution? What process technology and budgets were used?

3. What types of I/O have you designed? What were their size? Speed? Configuration? Voltage requirements?

Process technology? What package was used and how did you model the package/system?

What parasitic effects were considered?

4. What types of high speed CMOS circuits have you designed?

5. What transistor level design tools are you proficient with? What types of designs were they used on?

6. What products have you designed which have entered high volume production?

What was your role in the silicon evaluation/product ramp? What tools did you use?

7. If not into production, how far did you follow the design and why did not you see it into production?